

Parsa Mirtaheri

Department of Computer Science and Engineering
University of California, San Diego

✉ parsa@ucsd.edu
☎ +1 949-592-5937

EDUCATION	Ph.D. in Computer Science 01/23 - Expected 06/28 Department of Computer Science and Engineering University of California, San Diego Advisor: Misha Belkin <i>Selected Courses:</i> Data Systems for Machine Learning, Natural Language Processing, Probability Theory, Linear Algebra, Mathematics of Deep Learning, Unsupervised Learning B.Sc. in Computer Engineering 09/16 - 06/21 Department of Computer Engineering Sharif University of Technology, Tehran, Iran <i>Selected Courses:</i> Data Structures and Algorithms, Machine Learning, Stochastic Processes, Algorithmic Game Theory, Information Retrieval, Information Theory, Automata Theory High School Diploma in Mathematics and Physics 09/12 - 06/16 Allameh Helli 3 High School, Tehran, Iran Affiliated with the National Organization for the Development of Exceptional Talents
RESEARCH EXPERIENCE	Visiting Graduate Student, Simons Institute for the Theory of Computing at the University of California, Berkeley Fall 2024 Researching the reasoning capabilities of transformer-based models as part of the Modern Paradigms in Generalization program. Graduate Student Researcher, University of California, San Diego Researching the feature learning mechanism in neural networks and kernel machines that adapt to the low-dimensional structure of the task in Belkin Lab. Researcher, Sharif University of Technology Fall 2021 Worked on adversarial robustness and theoretical foundations of deep learning under the supervision of Prof. Rohban in the Robust and Interpretable Machine Learning lab. Remote Research Intern, Technical University of Munich Summer 2020 Computer Vision group of Prof. Navab. Research Intern, IST Austria Summer 2019 Game Theory & Computer-Aided Verification group of Prof. Chatterjee.
PUBLICATIONS	Parsa Mirtaheri, Ezra Edelman, Samy Jelassi, Eran Malach, Enric Boix-Adserà. <i>Let Me Think! A Long Chain of Thought Can Be Worth Exponentially Many Short Ones.</i> arXiv:2505.21825, under review, 2025. Parsa Mirtaheri, Mikhail Belkin. <i>Detecting Motivated Reasoning in the Internal Representations of Language Models.</i> Manuscript in preparation, 2025. Ali Shirali, Arash Nasr-Esfahany, Abdullah Omar Alomar, Parsa Mirtaheri, Rediet Abebe, Ariel D. Procaccia. <i>Direct Alignment with Heterogeneous Preferences.</i> arXiv:2502.16320, under review, 2025.

HONORS & AWARDS	• Bronze Medal in the ACM-ICPC World Finals, Porto, Portugal	2019
	Ranked 9 th among 135 teams of three chosen from a field of 52,709 contestants from 3,233 universities	
	• Silver Medal in 28 th International Olympiad in Informatics (IOI), Kazan, Russia	2016
	• Undergraduate Excellence Award, CAMP, Technical University of Munich	2020
	• Admission with a full scholarship to IST Summer Internship by OeAD-GmbH, Austria	2019
	• Gold Medal in 24 th Iran National Olympiad in Informatics, Tehran, Iran	2015
	• Silver Medal (ranked 2nd) in National Scientific Olympiad in Computer Engineering	2021
WORK EXPERIENCE	Research and Development Engineer, Map and Navigation Application	Summer 2017
	- Fall 2018	
	Developed the navigation service algorithm and the conflation algorithms for importing data from OpenStreetMap. During the first year, our application gained 200,000 users. I was one of five members of the team.	
	Course Designer, Quera	Summer 2020
	During my undergraduate industry internship at Quera, I was part of the team that designed two introductory courses on machine learning. About 8,000 students enrolled in the courses.	
TEACHING EXPERIENCE	Teaching Assistant, University of California, San Diego	
	• Online Learning, Instructor: Prof. Freund	2025
	• Mathematics of Deep Learning, Instructor: Prof. Belkin	2024
	• Machine Learning, Instructor: Prof. Belkin	2023
	Teaching Assistant, Sharif University of Technology	
	• Probability and Statistics, Instructor: Prof. Omidvar	2020
	• Discrete Structures, Instructor: Prof. Zarrabi	2020
	• Machine Learning, Instructor: Prof. Rohban	2020
	• Design of Algorithms, Instructor: Prof. Zarrabi	2019
	• Artificial Intelligence, Instructor: Prof. Rohban	2019
	• Data Structures and Algorithms, Instructor: Prof. Sharifi	2018
	Member of Iran National Olympiad in Informatics Scientific Committee	
	Head of Iran's IOI team selection exams scientific committee	2017, 2020
	Member of ACM-ICPC West Asia Regional Contest Scientific Committee	2019, 2020
	Lecturer, Allameh Helli 3 High School	2015-2016
VOLUNTEER EXPERIENCE	Elected Member of the Student Scientific Chapter (SCC) of CE Department	2018-2019
	Head of the Scientific & Technical Content Team of Sharif AIC	Winter 2018
	Executive Director of the 1 st Internship Seminar Series, <i>Interna</i>	Fall 2018
	Editor-in-chief of CE Department Collegiate Journal, <i>Rayanesh</i>	2017-2018
SKILLS	Programming Languages: C/C++, Python, Java, Pascal, SQL, Bash	
	Libraries: PyTorch, NumPy, Pandas, SciPy, Scikit-learn	
	Typesetting: L ^A T _E X	